

Student Document

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Q3 - GIT BRANCHING AND FILE UPDATES

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git branch
  course-search
* master
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ ls
backup/  course_info.txt  docs/  src/
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ nano course_info.txt # (updated the Project Name line)
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git add course_info.txt
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git commit -m "Update course information"
[master cdfdaeb] Update course information
1 file changed, 1 insertion(+), 1 deletion(-)
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git log --oneline --all --graph
* cdfdaeb (HEAD -> master) Update course information
* b45e4c5 Initial project setup
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ cat course_info.txt
Project Name: Task Scheduler with Priority Scheduling
Course Duration: 4 Years
Created By: Manvendra
Last Updated: 02-09-2026
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ □
```

Q4 - MERGING & CONFLICT RESOLUTION

PART A - Merge course-search

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git merge course-search
Updating c3f78831..c378831
Fast-forward
 src/course_search.txt | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 src/course_search.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ ls src
course_search.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git log --oneline --all --graph
* c378831 (HEAD -> master, course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup
```

----- PART B - Generate conflict and resolve -----

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git branch course-update
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git switch course-update
Switched to branch 'course-update'
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ nano course_info.txt # (edited the line: Project Name: Course Registration System)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ git add course_info.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ git commit -m "Update project name in course-update"
[course-update f5b3a72] Update project name in course-update
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ git switch master
Switched to branch 'master'
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ nano course_info.txt # (edited the line differently: Project Name: Online Course
Registration System)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git add course_info.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git commit -m "Update project name in master"
[master 3e8d1b9] Update project name in master
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
```

```
Unmerged paths:
  (use "git add <file>..." to mark resolution)
```

both modified: course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ nano course_info.txt # (manually resolved the conflict)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ cat course_info.txt
Project Name: Online Course Registration System
Course Duration: 4 Years
Created By: Manvendra
Last Updated: 02-09-2026
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ git add course_info.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ git commit -m "Resolve merge conflict in course information"
[master a7d9c41] Resolve merge conflict in course information
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git status
On branch master
nothing to commit, working tree clean
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git log --oneline --all --graph
* a7d9c41 (HEAD -> master) Resolve merge conflict in course information
|
| * f5b3a72 (course-update) Update project name in course-update
| * 3e8d1b9 Update project name in master
|/
* c378831 (course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup
```

Q1 - LINUX COMMANDS BASICS

1. Create and navigate to project directory

Manvendra@Manvendra MINGW64 ~

\$ mkdir Manvendra-GLA-CourseRegistration

Manvendra@Manvendra MINGW64 ~

\$ cd Manvendra-GLA-CourseRegistration

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ pwd

/c/Users/Manvendra/Manvendra-GLA-CourseRegistration

2. Create project structure

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ mkdir -p src docs backup

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ touch src/course_search.txt course_info.txt docs/README.md backup/backup_info.txt

3. List all files and directories

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ ls -R

.:
backup/ course_info.txt docs/ src/

./backup:
backup_info.txt

./docs:
README.md

./src:
course_search.txt

3 directories, 4 files

4. Check current location and contents (detailed)

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ ls -la

total 5
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 .
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:10 ..
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 backup
-rw-r--r-- 1 Manvendra 197609 0 Sep 02 10:15 course_info.txt
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 docs
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 src
total 5

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ █

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
##### Q5 - FINAL DEMONSTRATION #####
1. Complete Linux project structure
$ tree
.
|-- backup
|-- docs
|-- src
|   |-- course_search.txt
|   |-- course_info.txt
|   |-- 
|-- 
|-- 
|-- 

3 directories, 2 files
2. Git repository status
$ git status
On branch master
nothing to commit, working tree clean

3. Git history
$ git log --oneline --all --graph
* a7d9c41 (HEAD -> master) Resolve merge conflict in course information
| \
| * f5b3a72 (course-update) Update project name in course-update
| * | 3e8d1b9 Update project name in master
| /
* c378831 (course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup

4. Feature branches
$ git branch
  course-search
  course-update
* master

5. Successful feature merge (Final status)
$ git status
On branch master
nothing to commit, working tree clean

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ |
```

Q1 - LINUX COMMANDS BASICS

1. Create and navigate to project directory

Manvendra@Manvendra MINGW64 ~

\$ mkdir Manvendra-GLA-CourseRegistration

Manvendra@Manvendra MINGW64 ~

\$ cd Manvendra-GLA-CourseRegistration

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ pwd

/c/Users/Manvendra/Manvendra-GLA-CourseRegistration

2. Create project structure

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ mkdir -p src docs backup

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ touch src/course_search.txt course_info.txt docs/README.md backup/backup_info.txt

3. List all files and directories

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ ls -R

..
backup/ course_info.txt docs/ src/

./backup:
backup_info.txt

./docs:
README.md

./src:
course_search.txt

3 directories, 4 files

4. Check current location and contents (detailed)

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ ls -la

total 5
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 .
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:10 ..
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 backup
-rw-r--r-- 1 Manvendra 197609 0 Sep 02 10:15 course_info.txt
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 docs
drwxr-xr-x 1 Manvendra 197609 0 Sep 02 10:15 src
total 5

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration

\$ █

Q3 - GIT BRANCHING AND FILE UPDATES

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git branch
  course-search
* master
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ ls
backup/  course_info.txt  docs/  src/
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ nano course_info.txt # (updated the Project Name line)
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git add course_info.txt
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git commit -m "Update course information"
[master cdfdaeb] Update course information
1 file changed, 1 insertion(+), 1 deletion(-)
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ git log --oneline --all --graph
* cdfdaeb (HEAD -> master) Update course information
* b45e4c5 Initial project setup
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ cat course_info.txt
Project Name: Task Scheduler with Priority Scheduling
Course Duration: 4 Years
Created By: Manvendra
Last Updated: 02-09-2026
```

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

```
$ □
```

Q4 - MERGING & CONFLICT RESOLUTION

PART A - Merge course-search

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git merge course-search
Updating c3f78831..c378831
Fast-forward
 src/course_search.txt | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 src/course_search.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ ls src
course_search.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git log --oneline --all --graph
* c378831 (HEAD -> master, course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup
```

----- PART B - Generate conflict and resolve -----

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git branch course-update
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git switch course-update
Switched to branch 'course-update'
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ nano course_info.txt # (edited the line: Project Name: Course Registration System)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ git add course_info.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ git commit -m "Update project name in course-update"
[course-update f5b3a72] Update project name in course-update
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (course-update)
$ git switch master
Switched to branch 'master'
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ nano course_info.txt # (edited the line differently: Project Name: Online Course
Registration System)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git add course_info.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git commit -m "Update project name in master"
[master 3e8d1b9] Update project name in master
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ git status
On branch master
You have unmerged paths.
  (fix conflicts and run "git commit")
```

```
Unmerged paths:
  (use "git add <file>..." to mark resolution)
```

both modified: course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ nano course_info.txt # (manually resolved the conflict)
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ cat course_info.txt
Project Name: Online Course Registration System
Course Duration: 4 Years
Created By: Manvendra
Last Updated: 02-09-2026
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ git add course_info.txt
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master|MERGING)
$ git commit -m "Resolve merge conflict in course information"
[master a7d9c41] Resolve merge conflict in course information
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git status
On branch master
nothing to commit, working tree clean
```

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ git log --oneline --all --graph
* a7d9c41 (HEAD -> master) Resolve merge conflict in course information
|
| * f5b3a72 (course-update) Update project name in course-update
| * 3e8d1b9 Update project name in master
|/
* c378831 (course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup
```

Q2 - GIT REPOSITORY INITIALIZATION

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git init

Initialized empty Git repository in C:/Users/Manvendra/Manvendra-GLA-CourseRegistration/.git/

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git config user.name "Manvendra Arora"

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git config user.email "manvendra.arora@example.com"

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git add .

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git commit -m "Initial project setup"

[master b45e4c5] Initial project setup

2 files changed, 11 insertions(+)

create mode 100644 course_info.txt

create mode 100644 docs/README.md

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git status

On branch master

nothing to commit, working tree clean

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ git log --oneline

b45e4c5 (HEAD -> master) Initial project setup

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)

\$ █

```
Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
##### Q5 - FINAL DEMONSTRATION #####
1. Complete Linux project structure
$ tree
.
|-- backup
|-- docs
|-- src
|   |-- course_search.txt
|   |-- course_info.txt
|   |-- 
|-- 
|-- 
|-- 

3 directories, 2 files
2. Git repository status
$ git status
On branch master
nothing to commit, working tree clean

3. Git history
$ git log --oneline --all --graph
* a7d9c41 (HEAD -> master) Resolve merge conflict in course information
| \
| * f5b3a72 (course-update) Update project name in course-update
| * | 3e8d1b9 Update project name in master
| /
* c378831 (course-search) Add course search module
* cdfdaeb Update course information
* b45e4c5 Initial project setup

4. Feature branches
$ git branch
  course-search
  course-update
* master

5. Successful feature merge (Final status)
$ git status
On branch master
nothing to commit, working tree clean

Manvendra@Manvendra MINGW64 ~/Manvendra-GLA-CourseRegistration (master)
$ |
```